

each corner is measured to find the coordinates of the touch. Such panels weren't durable enough, so they flopped. More recently, we've seen optical imaging touchscreen technology. This uses cameras that sense usually far infrared light; when the light is interrupted by touch, the location can be calculated. Microsoft improved upon this technology and introduced the Surface, which you've doubtless heard about.

Microsoft's Surface

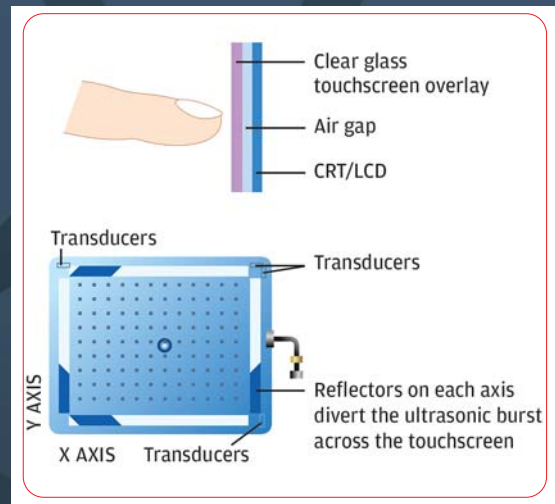
The Surface is a touchscreen computer that looks like a coffee table and allows for surface computing. Through "surface computing," you can seamlessly synchronise electronic devices that touch its surface. For example, the Surface allows you to connect a camera with a mobile. You can transfer images from your wireless-enabled camera to a Bluetooth-enabled mobile phone, with both on the Surface. On

ing of virtual objects like images and files across the table.

Misconceptions And Disadvantages

Misconceptions include the idea that grease will spoil the surface of a touchscreen. Not true; touchscreens with glass surfaces can resist dirt, grease, moisture, and also household cleaning agents. Another misconception is that touchscreen displays gets washed out when viewed under sunlight. This has nothing to do with the touchscreen in itself; visibility depends on the quality and contrast of the display.

With improvement in technology of touchscreens, precise pointing is also possible. Having said that, touchscreens have their share of disadvantages. People with fat fingers may mis-hit keys and find it tough to navigate. One needs to be careful while handling mobile devices with touchscreens because the screen gets damaged when kept in a pocket



Acoustic touchscreens—accurate, and good for big displays

provides plenty of tactile feedback. But recent demonstrations of the TouchFLO mobile phone from High Tech Computer Corp (who design, manufacture, and market smartphones and PDA-phones) have shown that the combination of a finger and a UI designed for the finger can make for data input as fast as with a keyboard.

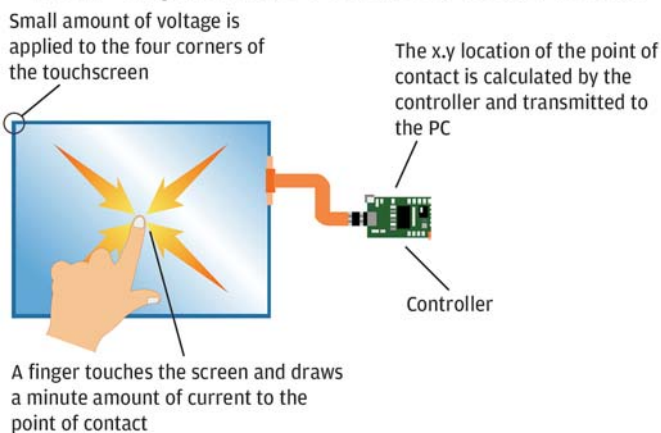
And There's More!

Many large companies—Microsoft and Apple included—have gotten on the touchscreen bandwagon, multi-touchscreens in particular. After the iPhone, rumour has it that Apple will soon come up with a tablet PC wherein multiple fingers (up to 15 simultaneous touches!) can be tracked. It's further been said that Apple has patented an "Integrated Sensing Display" wherein display elements are integrated with image sensing elements. (Think of "image sensing elements" as micro-cameras.) If this were to be put to use, you could have a single device—that looked like a monitor—for videoconferencing, wherein the monitor would be the camera!

As you've been noticing, keypads on mobile devices are getting cramped—with more and more buttons for launching all the various applications. With the number of buttons that can fit on a mobile device reaching its limit, it might not be far-fetched to predict the gradual death of the button, giving way to the touchscreen. ■

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How Capacitive Touchscreens Work



Capacitive touchscreens – clear screens, but at a high cost

placing the wireless-enabled camera on the tabletop, pictures are wirelessly transferred to the hard drive of the Surface. Using your fingers, you can fiddle with and pinch the images to rotate, crop, or resize them. To transfer images, you put your finger on the Surface and *just drag* the images to your Bluetooth-enabled phone!

Thanks to multi-touch, the Surface can accept input from multiple fingers and multiple users simultaneously and allow complex gestures like grabbing, stretching, swivelling, and slid-

ing with coins and keys. Touchscreens are complex items, and unlike keypads, there are many things that can go wrong: deteriorating response, unruly calibration, and dead pixels.

Touchscreen tech has its critics: of all people, Nokia's general manager for mobile devices, Antti Vasara, claims people don't like touchscreens and prefer keyboards and joysticks. He says they wouldn't like to use an interface designed for a mouse—such as a regular monitor—as a touchscreen, and that nothing beats a keyboard, which